Cooperative Fleet Lane Changing With Multi-Group Splitting and Merging Based on Vehicular Sensor Networks

Lien-Wu Chen, *Senior Member, IEEE*, and Yu-An Shi

***Abstract*—In this paper, we propose a cooperative fleet lane changing (CFLC) framework based on vehicular sensor networks, which enables multiple groups of fleet vehicles splitting and merging. The proposed CFLC framework determines the lane-changing space in the target lane, performs real-time splitting of fleet vehicles, coordinates the speeds of fleet vehicles and non-fleet vehicles, and merges multiple groups of fleet vehicles. In CFLC, we design the dynamic programming algorithm to determine the optimal fleet lane-changing space, and split fleet vehicles into multiple groups according to the optimal lane-changing space. As there are no sufficient inter-vehicle gaps for fleet group lane changing, hybrid speed control is explored to adjust both the fleet vehicle speeds and the speeds of vehicles in the target lane to avoid unnecessary acceleration/deceleration. Moreover, efficient merging of multiple fleet groups is designed to deal with the separation of fleet vehicles for avoiding increased travel time and fuel/energy consumption. According to our review of relevant research, this is the first cooperative fleet lane changing framework that supports multi-group vehicle splitting and merging. Simulation results show that our framework outperforms existing lane-changing methods and can minimize the speed variations of fleet vehicles and non-fleet vehicles for improving traffic flow and reducing fuel/energy wastage. In particular, with a fleet of 10 vehicles at 110 km/hr, average speed variations for changing lanes are decreased by more than 14 m/s and improved by more than 45%.**

***Index Terms*—Cooperative fleet lane changing, dynamic programming, fleet splitting and merging, hybrid speed control, vehicular sensor network.**

I. Introduction

With the widespread adoption and advancement of smart vehicles, information and communication technologies are applied in traffic signals, roads, and vehicles. Vehicular Sensor Networks (VSNs), which is a combination of Wireless Sensor Networks (WSNs) and Vehicular Ad Hoc Networks (VANETs), have emerged to improve traffic safety and driving efficiency through vehicular communications, including Autonomous Emergency Braking, Lane Departure Warning, Cooperative Adaptive Cruise Control, etc. [1]. Developed and developing countries have invested significant funds and efforts in traffic-related improvement, particularly focusing on fuel/energy consumption and traffic quality [2]. Fuel/energy consumption efforts focus on reducing unnecessary acceleration/deceleration and optimizing route planning [3]. Traffic quality is mainly targeted toward smart cities, integrating road design with Internet of Vehicles communications including Vehicle-to-Device (V2D), Vehicle-to-Vehicle (V2V), Vehicle-to-Grid (V2G), and Vehicle-to-Everything (V2X) [4], [5]. Research works on road design includes intelligent median islands [6], communications between traffic lights and vehicles [7], lane-free autonomous intersection management [8], and enhanced public facilities along roads [9].

To enhance driving efficiency, the stability of traffic flow and string stability are notably impacted by the acceleration and deceleration patterns of vehicles. Excessive acceleration and braking behaviors can lead to fuel wastage and traffic congestion [10]. Research focusing on enhancing vehicle fuel economy indicates that the disparity in fuel consumption between the least efficient and most efficient drivers is approximately 70% [11]. Specifically, mitigating unnecessary acceleration and deceleration among drivers can reduce the costs associated with traffic congestion, including time and fuel wastage [12].

Existing studies in cooperative adaptive cruise control (CACC) aim to improve traffic flow [13], [14], [15], [16], [17] and reduce fuel/energy consumption [18], [19], [20], [21], [22], [23]. Few works are dedicated to fleet vehicles for cooperative fleet cruise control, whereas most works focus on controlling individual vehicles. Fleet management systems with CACC are designed to achieve intelligent transportation [24], [25] and address energy/environmental concerns [26], [27]. To alleviate congestion, prevent accidents, and/or increase throughput during lane changing, state-of-the-art methods utilize vehicle trajectories for lane changing [28], employ speed control for lane changing [29], coordinate vehicle lane changing [30], [31], [32], [33], [34], [35], [36], control lane-changing behaviors of fleet vehicles [37], platooning management in mixed traffic flow [38], lane change with right-of-way assignment [39], cooperative-driving control for wireless charging [40], and negotiation-based right-of-way assignment [41].

Similarly, most lane-changing strategies are designed for individual vehicles, which change lanes in a short time and

Received 11 April 2024; revised 14 November 2024 and 16 May 2025; accepted 4 July 2025. Date of publication 6 August 2025; date of current version 21 October 2025. This work was supported in part by the National Science and Technology Council, Taiwan, under Grant 112-2221-E-035-032 and Grant 113-2221-E-035-073-MY3. The Associate Editor for this article was C. Ding. *(Corresponding author: Lien-Wu Chen.)*

The authors are with the Department of Information Engineering and Computer Science, Feng Chia University, Taichung 407, Taiwan (e-mail: lwuchen@fcu.edu.tw).

Digital Object Identifier 10.1109/TITS.2025.3593961

1524-9050 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON OF PRIOR WORKS FOR LANE CHANGING [28], [29], [30], [31], [32], [33], [34], [35], [36], [37] AND OUR CFLC FRAMEWORK

Features	Individual Lane Changing	Cooperative Lane Changing	Fleet Lane Changing	Fleet Vehicle Splitting	Fleet Vehicle Merging	Lane-Changing Space Optimization
SMPC [28]	√					
VSL-LCC [29]	√					
PSO-LCM [30]	√	√				
CTM-LCC [31]	√	√				
CDLC [32]	√	√				
CLC-DMP [33]	√	√				
CLCS-IV [34]	√	√				
CSDM [35]	√	√				
CLCS [36]	√	√				
CFCC [37]	√	√	√			
Our Framework	√	√	√	√	√	√

reduce fuel/energy consumption, but require waiting for the dedicated space larger than the entire fleet to perform lane changing. With high traffic flow in the target lane, it is challenging to acquire a sufficient space for the fleet to change lanes efficiently.

In this work, we propose a cooperative fleet lane changing (CFLC) framework based on VSNs, which enables the splitting of fleet vehicles and the merging of multiple groups of fleet vehicles. The proposed framework explores a dynamic programming algorithm to determine the optimal aggregated fleet lane-changing space consisting of consecutive inter-vehicle gaps in the target lane, where the inter-vehicle gap is the space between a vehicle and its immediate front vehicle in the same lane. Next, based on the determined optimal lane-changing space, fleet vehicles are segmented into multiple groups for lane changing. If necessary, hybrid speed control is employed to coordinate the speeds of fleet vehicles in the current lane and non-fleet vehicles in the target lane (i.e., target-lane vehicles). Finally, the segmented group of fleet vehicles is efficiently merged to avoid the fleet from separating, which can minimize unnecessary acceleration/deceleration and reduce additional fuel/energy consumption. According to our review of relevant research, this is the first cooperative fleet lane changing framework that enables multi-group vehicle splitting and merging.

The contributions of our framework are four-fold. First, the optimal lane-changing space is determined to minimize the number of fleet groups required for efficient maneuvering, which can alleviate unnecessary traffic disruption. Second, the speeds and acceleration/deceleration of both fleet vehicles and vehicles in the target lane are effectively coordinated, which can reduce speed variations during lane-changing and avoid significant speed fluctuations and potential traffic instability. Third, the splitting of fleet vehicles is performed based on the determined optimal lane-changing space to allow each group to seamlessly change lanes into the designated inter-vehicle gaps, which can minimize inefficient lane usage and increased congestion. Finally, the feasible space required for merging fleet groups is planned with minimal speed variations based on the number of fleet groups and their respective positions, which can avoid pronounced speed discrepancies and reduced traffic flow efficiency.

The rest of this paper is organized as follows. Section II discusses existing works. Section III defines our cooperative fleet lane changing problem. Section IV presents our framework to solve this problem. Simulation results are discussed in Section V. Finally, Section VI concludes the paper.

II. RELATED WORK

Tab. I shows the comparisons of existing methods for individual lane changing [28], [29], cooperative lane changing [30], [31], [32], [33], [34], [35], [36], fleet lane changing [37], and our CFLC framework. CFLC offers the most comprehensive solution to multi-group fleet lane changing according to whether (1) lane changing can be safely performed by an individual vehicle through vehicular communications (i.e., Individual Lane Changing), (2) individual vehicles in the current and target lanes can cooperate in speed control to ensure sufficient space for lane changing (i.e., Cooperative Lane Changing), (3) lane changing can be performed by an entire fleet to enhance driving efficiency (i.e., Fleet Lane Changing), (4) fleet vehicles can be divided into multiple groups as there is no single sufficient inter-vehicle gap for the entire fleet to change lanes (i.e., Fleet Vehicle Splitting), (5) fleet groups can be effectively merged after passing through the incident point (i.e., Fleet Vehicle Merging), and (6) the optimal lane-changing space can be determined to minimize the speed variation and reduce fuel/energy consumption (i.e., Lane-Changing Space Optimization).

Reference [28] designed a control method and motion planning for lane changing in autonomous vehicles based on probabilistic and deterministic prediction. The designed control method is based on Stochastic Model Predictive Control (SMPC), which determines the inputs needed for control driving performance and prevents vehicle collisions. The vehicle model uses the image sensor to identify the curvature of the road and calculates the translational inertia based on the vehicle weight, enhancing the accuracy of path prediction. During lane changing, the method calculates a safety space for vehicles. However, SMPC is designed for a single vehicle using the camera to analyze the trajectories of surrounding vehicles, and does not facilitate prediction for a fleet of vehicles to achieve multi-group fleet lane changing.

Reference [29] presented an approach for evaluating and analyzing lane-changing control strategies and variable speed limits on highway traffic. The presented approach determines lane-changing strategies based on traffic demand and bottleneck formation, suggesting strategies to alleviate the reduction

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

in throughput at bottleneck locations. The approach employs a nonlinear variable speed limit (VSL) speed controller to ensure the maximum flow rate at bottleneck locations. In the VSL control, the highway is divided into a number of segments, and the traffic density of each segment is calculated. VSL control is then applied to mitigate traffic bottlenecks. However, the presented approach focuses solely on variable speed limits and lane-changing strategies to alleviate traffic bottlenecks. It does not leverage flexible multi-vehicle cooperative speed control, limiting its effectiveness in alleviating traffic bottlenecks during lane changing.

Reference [30] used Particle Swarm Optimization (PSO) in the context of lane changing for autonomous vehicles with the PSO Lane Change Model (PSO-LCM). This model offers rules for lane changing, merging vehicles, and collision avoidance for autonomous vehicles. It employs an adaptive approach to enhance merging efficiency and provides a lane-changing model capable of learning merging strategies with minimal collisions. While the proposed method utilizes communication for cooperative lane merging strategies, it does not address the simultaneous merging of multiple vehicles to an entire fleet into the original lane.

Reference [31] proposed a traffic congestion control strategy to alleviate freeway lane-drop bottlenecks, which utilizes both variable speed limit control and lane changing control, optimizing the traffic network performance through Genetic Algorithm under a connected and automated vehicle environment. Multi-class Cell Transmission Model, commonly used for traffic flow prediction, is applied to predict traffic conditions on a given highway segment. However, this strategy employs speed limit control to alleviate lane-drop bottlenecks and does not incorporate flexible speed control to address lane-changing behaviors.

Reference [32] developed a Cooperative Driving and Lane Changing (CDLC) model within the proximity of traffic signals, incorporating vehicular communications. The CDLC model aims to decrease the frequency of vehicle stops and overall travel time by preplanning vehicle trajectories in the vicinity of traffic signals. Moreover, CDLC enhances safety and comfort during the lane-changing process by planning cooperative lane-changing maneuvers based on the interactions between the host vehicle in the original lane and surrounding vehicles in the target lane. However, fleet lane changing and multi-group fleet splitting/merging are not addressed in the developed model.

Reference [33] presented a decentralized vehicle motion planning approach for cooperative lane changing, which redistributes a sufficiently sparse distribution of vehicles and ensures all vehicles connected to autonomous vehicles can simultaneously change lanes without collisions. When a formation reset is necessary, the presented method maintains consistent performance under constraints while facilitating lane changing. However, the goal is primarily on avoiding collisions between vehicle trajectories during lane changing without addressing spatial allocation for a fleet and multi-group fleet lane changing.

Reference [34] designed an intelligent lane-changing strategy for vehicles, encompassing both individual lane changing and cooperative lane changing involving multiple vehicles. The designed strategy assesses the feasibility of individual lane changing, considering potential mutual interference in lane changing decisions among vehicles. Using game theory and a matrix-based approach, a cooperative lane change decision model is developed to address the anticipated mutual influence in lane changing decisions among multiple vehicles. While this strategy employs communication for cooperative lane changing, it does not address the fleet lane changing with multi-group splitting and merging.

Reference [35] presented a cooperative space distribution method (CSDM) for vehicle lane changing, specifically focused on autonomous vehicles changing lanes on highways, which not only enhances lane throughput but also coordinates the dynamics of autonomous vehicles upstream in the lane. By redistributing vehicles on highway lane segments to other lanes, the presented CSDM method ensures that a sufficient safety distance is maintained both in front and behind when changing lanes. During lane changing, CSDM selects the appropriate gap between vehicles based on the vehicle position to reduce the average travel time for all vehicles and ensuring safety driving conditions. Although the collaboration for vehicles during lane changing is achieved, CSDM does not address simultaneous lane changing by multiple vehicles/groups in a fleet.

Reference [36] proposed a cooperative lane changing strategy (CLCS) based on vehicular communications, which utilizes cooperative acceleration and deceleration of vehicles to address road safety and congestion issues. The proposed CLCS strategy controls obstructed vehicles during lane changing, coordinating with the acceleration and deceleration of vehicles in other lanes to rapidly navigate through congested areas. CLCS reduces unnecessary lane changes and speed adjustments to enhance road safety. Although CLCS facilitates cooperative vehicle lane changing, it does not consider lane changing for a fleet of vehicles.

Reference [37] developed cooperative fleet cruise control (CFCC) based on cooperative adaptive cruise control in a fleet-driving manner, which reduces the time taken for vehicles to change lanes and enables a fast return to the original speed after emergency braking. CFCC improves fleet recovery efficiency and driving safety during emergency events. Although CFCC involves a strategy for fleet lane changing, it does not consider the aggregated lane changing space of inter-vehicle gaps for multiple fleet groups, as it requires a single sufficiently large space for the fleet to change lanes.

III. System Model

Fig. 1 shows the system architecture of CFLC. Each vehicle is equipped with an Onboard Unit (OBU) device. The OBU includes the following functionalities: Safety Distance Calculation to estimate a safety following distance to prevent rear-end collisions; Following Risk Assessment to estimate the risk of rear-end collisions based on the distance to the preceding vehicle; Lane Changing Coordination to coordinate lane-changing behaviors based on the relative positions and speeds of fleet vehicles and target-lane vehicles; and Wireless Access in Vehicular Environment/Dedicated Short

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

Fig. 1. System architecture of CFLC.

Range Communication (WAVE/DSRC) to support vehicular communications between vehicles.

In the 5.9 GHz frequency range, WAVE/DSRC divides the 75 MHz bandwidth into six service channels and one control channel. These channels are assigned dedicated numbers of 172, 174, $\cdots$, and 184. The control channel is designated as channel 178, while channel 172 serves as a common safety channel among vehicles. The WAVE/DSRC network interface periodically switches between the control channel and service channels every 50 milliseconds (i.e., broadcast rate of 10 Hz in the control channel). The allocation of the highest priority can be used to minimize the transmission delay and packet loss rate for CFLC lane-changing messages. Although Cellular-V2X offers broader coverage, scalability, and future-proofing with 5G integration, making it suitable for a wide range of use cases, it depends on cellular networks for specific services and faces interoperability challenges. In contrast, WAVE/DSRC provides highly reliable, low-latency communication in a dedicated spectrum band, making it ideal for vehicular safety applications.

Through WAVE/DSRC, vehicles can obtain and share various information from other ones, including the current vehicle speed (acquired through the OBD-II interface), vehicle position (determined using lane-level positioning technology to identify the current lane position), braking performance (the maximum braking deceleration of the vehicle), acceleration, deceleration, and the durations of acceleration and deceleration. The Global Positioning System (GPS) is a widely used satellite positioning technology in daily life, which is capable of calculating the direction, speed, travel time, and trajectory of a vehicle. Millimeter-Wave Radar (MWR) utilizes wireless antennas to transmit millimeter-wave signals and, by receiving the reflected millimeter-wave signals, calculates the current position, distance to the target, angles, and relative velocity.

In CFLC, fleet vehicles and non-fleet vehicles drive on the road and employ IEEE 802.11p network interfaces operating on WAVE/DSRC frequency bands. The fleet vehicles communicate with non-fleet vehicles through V2V communication, transmitting vehicle-related information to achieve improved fuel/energy efficiency and higher lane space utilization. Before fleet lane changing, CFLC lane-changing messages are transmitted with the highest priority to minimize the transmission delay and packet loss rate. During fleet lane changing, the millimeter-wave radar is used to keep detecting the unexpected vehicles/objects next to the fleet vehicle for handling emergency situations and avoiding side-impact collisions. After fleet lane changing, the car-following risk is estimated to keep the sufficient safety following distance for avoiding rear-end collisions between the fleet vehicle and its immediate front vehicle in the target lane.

To avoid unnecessary acceleration/deceleration and fuel/energy wastage for fleet lane changing due to an incident (e.g., road construction, traffic accident, lane reduction, etc.), we investigate the following four research issues:

1. Lane-Changing Space Planning: How can we determine the optimal fleet lane-changing space required by multiple groups of fleet vehicles to move into the target lane?
2. Hybrid Speed Control: How can we coordinate the speeds of fleet vehicles and target-lane vehicles with minimal acceleration/deceleration to obtain the necessary space for lane changing?
3. Multiple Group Splitting: How can we perform the splitting of fleet vehicles based on the determined optimal lane-changing space, ensuring each group can successfully change lanes into the designated spaces?
4. Fleet Vehicle Merging: How can we merge each group into its original fleet with minimal speed variations for fleet vehicles and target-lane vehicles after passing the incident location?

IV. Cooperative Fleet Lane Changing

Fig. 2 shows the lane-changing process for a single-group fleet in our previous work [37]. The fleet first checks if there

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

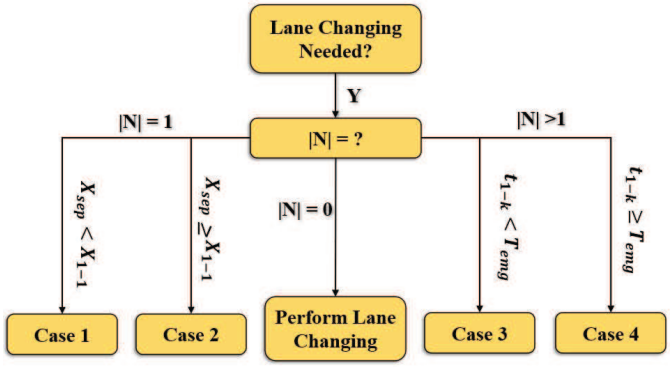

Fig. 2. Operation flow of single-group fleet lane changing.

is any vehicle in the target lane (i.e., target-lane vehicle). When there is no vehicle (i.e., $|N| = 0$), lane changing can be performed directly. As the target lane contains one vehicle (i.e., $|N| = 1$), the decision to either accelerate (Case 1) or decelerate (Case 2) the fleet vehicles is made based on the position of that target-lane vehicle (i.e., X_{1-1}) and the position of the lead vehicle in the fleet (i.e., X_{sep}) to expand the lane-changing space.

As there are more than one vehicle in the target lane (i.e., $|N| > 1$), the fleet calculates the expected time T_{emg} to reach the incident location (i.e., accident location or construction site that the fleet has to change lanes for passing through) based on the distance between fleet vehicles and the incident location and the speed of the lead fleet vehicle. It then determines whether the target-lane vehicle (i.e., C_{1-k}) can expand the space required for fleet vehicles to change lanes within the time t_{1-k} to move outside the required fleet lane-changing space spent by C_{1-k}. If T_{emg} is larger than t_{1-k}, this indicates that the target-lane vehicle has sufficient time to expand space for fleet vehicles to change lanes (Case 3). In contrast, if T_{emg} is less than or equal to t_{1-k}, fleet vehicles must perform a braking maneuver and wait until there is sufficient space in the target lane to change lanes (Case 4).

In this work, we further design a cooperative fleet lane changing framework that enables multiple group splitting and merging. The fleet vehicles drive on a multi-lane road and encounter an incident on the current lane, and adjacent lanes do not have sufficient space for the entire fleet to perform lane changing. In such cases, we split fleet vehicles into multiple groups, allowing lane changing without the need for a large space to accommodate the entire fleet.

Fig. 3 shows the process of multi-group cooperative fleet lane changing. If there is no vehicle in the target lane, the fleet performs lane changing directly (i.e., Perform Lang Changing). As the target lane contains one or more vehicles, if T_{emg} is larger than t_{1-k}, single-group fleet lane changing is performed (i.e., Perform Cooperative Lane Changing); otherwise, multi-group fleet lane changing has to be performed (i.e., Lane-Changing Space Planning). After inter-vehicle gaps are aggregated, as there is a sufficient aggregated space of gaps between vehicles in the target lane (i.e., Successful Planning), the fleet vehicles are split into multiple groups. Group Speed Control is performed to reduce the distance between group vehicles and the planned aggregated space, which enables group vehicles to change lanes to the designated inter-vehicle gaps in the target lane (i.e., Multiple Groups Splitting). As no aggregated space is sufficient (i.e., Unsuccessful Planning), Hybrid Speed Control is performed for both fleet vehicles and target-lane vehicles, where aggregated spaces are estimated to find the lane-changing space with minimal speed variations. In particular, specific inter-vehicle gaps are expanded to allow all fleet vehicles to perform lane changing as necessary (i.e., Cooperative Fleet Splitting). To enhance fleet driving efficiency, all fleet groups pass the incident location and merge back into the original fleet (i.e., Fleet Vehicles Merging). Tab. II lists the summary of notations used in the following multi-group fleet lane changing.

TABLE II
SUMMARY OF NOTATIONS

Notation	Meaning
b_{1-i}	Length of vehicle C_{1-i} in the target lane
X_{1-i}	Position of vehicle C_{1-i} in the target lane
A_i	Required lane-changing space for fleet vehicle C_{2-i}
D_i	Inter-vehicle gap between C_{1-i} and $C_{1-(i+1)}$
$Loc(D_i)$	Starting location of inter-vehicle gap D_i
$D(i,j)$	Aggregated space consisting of D_i, D_{i+1}, $\cdots$, and D_j
$N(i,j)$	Number of inter-vehicle gaps from D_i to D_j
$P_{D(i,j)}$	Spatial efficiency of the aggregated space $D(i,j)$
N_{gap}	Total number of inter-vehicle gaps in the searching range
S_i	Speed of vehicle C_i
E_i	Safety following distance for vehicle C_i
a_i^{max}	Maximum deceleration of vehicle C_i
δ	Average driver's reaction time
t^{cha}	The time taken for lane changing
h	Space required for lane changing of the last fleet group
Q_{fleet}	Speed variation of fleet vehicles
Q_{targe}	Speed variation of target-lane vehicles
Q	Speed variation of affected vehicles ($= Q_{fleet} + Q_{targe}$)
C_{j-k}	The k-th vehicle in lane j
E_{j-k}	Safety following distance of vehicle C_{j-k}
g_{j-k}	The k-th flee group in lane j
$a_{j-g_k}^{dec}$	Deceleration of fleet group g_{j-k}
$t_{j-g_k}^{dec}$	Deceleration period of fleet group g_{j-k}

A. Lane-Changing Space Planning

To plan the optimal space for fleet lane changing, the fleet vehicles search for available inter-vehicle gaps in the adjacent target lane (i.e., Lane 1). Assume that the length of the space occupied by fleet vehicles in the current lane (i.e., Lane 2) is $L_F = X_{2-1} - X_{2-n} + b_{2-n}$, where X_{2-1} is the x-coordinate of the lead fleet vehicle, X_{2-n} is the x-coordinate of the last fleet vehicle, and b_{2-n} is the length of the last fleet vehicle. The searching range in the target lane is set to α times of the fleet length, which consists of inter-vehicle gaps between target-lane vehicles (i.e., C_{1-1}, C_{1-2}, $\cdots$, and C_{1-m}, where m is the total number of target-lane vehicles in the searching range). When no sufficient aggregated lane-changing space of inter-vehicle gaps is found within the range of $\alpha \times L_F$ from the location of the lead vehicle in the fleet, the hybrid speed control method is designed to expand the aggregated lane-changing space in the target lane, allowing the fleet to perform lane changing successfully.

Our lane-changing space planning method consists of the following four steps. In the first step, we calculate the gaps

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

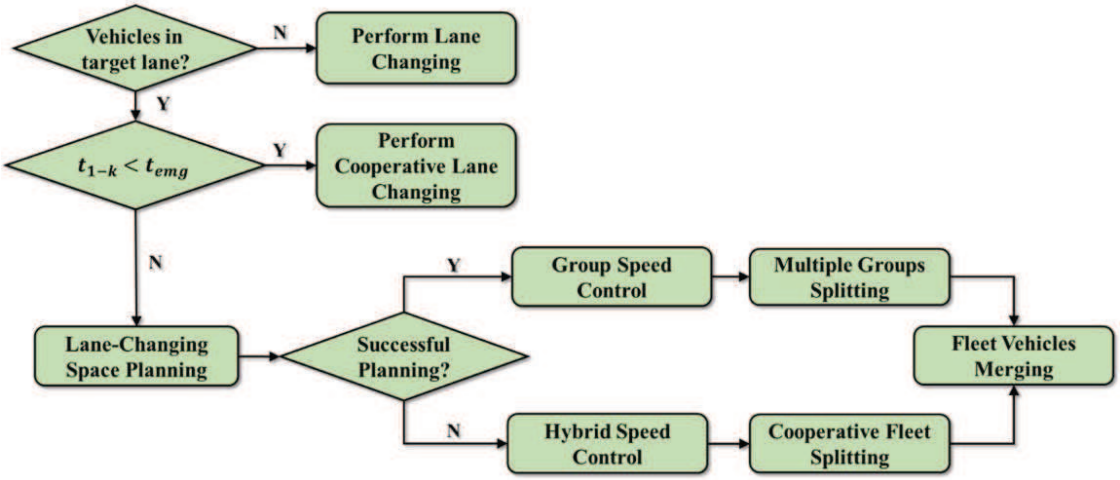

Fig. 3. Operation flow of multi-group fleet lane changing.

between consecutive vehicles in the target lane, resulting in available inter-vehicle gaps, denoted as D_i. In the second step, the calculated inter-vehicle gaps are sorted based on the position of the lead vehicle in the fleet, with D_1 representing the inter-vehicle gap closest to the lead vehicle. The third step involves selecting consecutive inter-vehicle gaps based on their positions to form aggregated lane-changing spaces. Finally, the optimal lane-changing space is planned as the sufficient aggregated space with the minimum number of consecutive inter-vehicle gaps

First, the gaps between vehicles in the target lane to be aggregated as lane-changing spaces for the fleet groups are calculated as

$$D_i = |X_{1-i} - X_{1-(i+1)}| - b_{1-i}, \tag{1}$$

where X_{1-i} is the position of the i-th vehicle (i.e., C_{1-i}) in the target lane, $X_{1-(i+1)}$ is the position of the $(i+1)$-th vehicle (i.e., $C_{1-(i+1)}$) in the target lane, b_{1-i} is the length of C_{1-i}, and D_i is the gap length between C_{1-i} and $C_{1-(i+1)}$.

Next, the calculated inter-vehicle gaps are sorted by gap sizes. To avoid dividing the fleet into too many groups when changing lanes, consecutive large inter-vehicle gaps are selected as the aggregated lane-changing space. In our designed dynamic programming algorithm, space searching and sorting for the calculated inter-vehicle gaps are performed to obtain the optimal lane-changing space. First, an empty 2D matrix $D^{N\times N}$ with the same length and width is adopted, where N is the total number of available inter-vehicle gaps in the searching ranging. In particular, the aggregated spaces of consecutive inter-vehicle gaps needed for lane changing are calculated by using the following recurrence relation

$$D(i,j) = \begin{cases} D(i,j-1) + D(j,j) & if\, i < j \\ D(i-1,j) + D(i,i) & if\, i > j \\ D_i & if\, i = j \end{cases}. \tag{2}$$

Fig. 4 shows an example of consecutive inter-vehicle gaps and aggregated spaces, where blue cars are fleet vehicles (with the fleet length of L_F) encountering an incident (from the distance of D_{emg} away) in the current lane (i.e., lane 2) and red cars

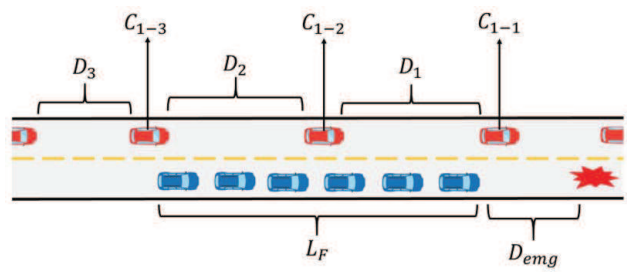

Fig. 4. Consecutive inter-vehicle gaps.

TABLE III

AGGREGATED SPACES OF CONSECUTIVE INTER-VEHICLE GAPS (METERS)

$D(i,j)$	1	2	3	4	5	6	7	8	9	10
1	**6**	13	23	28	37	48	57	69	83	99
2	13	**7**	17	22	31	42	51	63	77	93
3	23	17	**10**	15	24	35	44	56	70	86
4	28	22	15	**5**	14	25	34	46	60	76
5	37	31	24	14	**9**	20	29	41	55	71
6	48	42	35	25	20	**11**	20	32	46	62
7	57	51	44	34	29	20	**9**	21	35	51
8	69	63	56	46	41	32	21	**12**	26	42
9	83	77	70	60	55	46	35	26	**14**	30
10	99	93	86	76	71	62	51	42	30	**16**

are non-fleet vehicles in the target lane (i.e., lane 1). $D(1,3)$ is the aggregated space of consecutive inter-vehicle gaps, which is calculated as

$$D(1,3) = D(1,2) + D(3,3) = D(1,1) + D(2,2) + D_3. \tag{3}$$

The inter-vehicle gaps less than the length of a vehicle plus the required safety distance are excluded, and the eligible inter-vehicle gaps D_i are the default values in the diagonal positions of $D^{N\times N}$ (i.e., $D(i,j) = D_i$ if $i = j$). The aggregated lane-changing spaces can be calculated based on the eligible gaps, as listed in Tab. III. For instance, starting from the third inter-vehicle gap $D(3,3)$ of 10 meters, the aggregated space $D(2,3)$ is 17 meters (i.e., $D(2,3) = D(2,2) + D(3,3)$), where the aggregated space from third gap to the tenth gap is 86 meters (i.e., $D(3,10)$). Next, based on the recursively summed

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CANDIDATES OF AGGREGATED LANE-CHANGING SPACES (METERS)

D_i	1	2	3	4	5	6
6	57	69	83	99		
7	63	77	93			
10	56	70	86			
5	60	76				
9	55	71				
11	62					
9	57					
12	56	63	69			
14	55	60	70	77	83	
16	62	71	76	86	93	99

TABLE V
SORTED LANE-CHANGING SPACE CANDIDATES (METERS)

S	D_i	F
5	9	55
5	14	55
5	11	62
5	16	62
6	10	56
6	12	56
6	5	60
6	14	60
6	9	71
6	16	71

inter-vehicle gaps, we select the aggregated spaces larger than or equal to the total fleet length as candidates.

Suppose that the fleet length is 52 meters, Tab. IV lists the aggregated spaces larger than or equal to the fleet length with different number of inter-vehicle gaps. Finally, these aggregated spaces are sorted by their numbers of inter-vehicle gaps in an increasing order. The minimum number of inter-vehicle gaps where the aggregated space matches the fleet length is selected as the optimal lane-changing space. Tab. V lists the sorted result with the number of gaps (S), starting gap (D_i), and aggregated size (F). Note that when two or more aggregated spaces have the minimum number of inter-vehicle gaps, the aggregated space closest to the lead vehicle in the fleet is selected.

Next, we analyze the time complexity of the lane-changing space optimization algorithm. In the worst case, we are solving each subproblem only once. For lane-changing space optimization, there are $N \times N$ different subproblems (i.e., $N \times N$ aggregated spaces of consecutive inter-vehicle gaps), where N is the total number of available inter-vehicle gaps in the searching ranging. Since the number of aggregated spaces larger than or equal to the fleet length is N^2 at most, the running time of sorting these aggregated spaces by their numbers of inter-vehicle gaps in an increasing order is $O(N^2 lg(N^2))$ at most. Thus, the time complexity is $O(N^2 + 2N^2 lgN)$.

Algorithm 1 gives the pseudocode of the lane-changing space optimization algorithm, where N_{gap} is the total number of inter-vehicle gaps, D_i^{gap} is the gap between vehicles i and $i+1$, L_F is the fleet length, S_{opt} is the optimal lane-changing space, S_{start} is the first gap of S_{opt}, and S_{end} is the last gap of S_{opt}. Through lane-changing space planning, the optimal lane-changing space can be determined with the minimum number of fleet groups, which which alleviates unnecessary traffic disruption.

Algorithm 1 Lane-Changing Space Optimization

```
Input: N_gap, D^gap, L_F
Result: S_opt, S_start, S_end
begin
    for i ← 1 to N_gap do
        D(i, i) = D_i^gap
    for i ← 1 to N_gap do
        for j ← i + 1 to N_gap do
            D(i, j) = D(i, j − 1) + D(j, j)
        for j ← i − 1 to 0 do
            D(i, j) = D(i − 1, j) + D(i, i)
    S_start = 1
    S_end = N_gap
    for i ← 1 to N_gap do
        for j ← 1 to N_gap do
            if D(i, j) > L_F & |S_start − S_end| > |i − j|
            then
                S_opt = D(i, j)
                if i < j then
                    S_start = i
                    S_end = j
                else
                    S_start = j
                    S_end = i
    return S_opt, S_start, S_end
```

B. Dynamic Fleet Group Splitting

In this subsection, we calculate the optimal safety following distance between vehicles using the car-following risk estimation in [37]. When the car-following risk value exceeds 1.0, there is a rear-end collision between vehicles as emergency braking. The minimum safety following distance E_i for vehicle C_i to the immediate front vehicle C_{i-1} is calculated using the car-following risk value of 1.0 as

$$E_i = \frac{(S_i)^2 - (S_{i-1})^2}{2 \cdot a_i^{max}} + S_i \cdot \delta, \tag{4}$$

where S_i and S_{i-1} are the speeds of vehicles C_i and C_{i-1}, respectively, a_i^{max} is the maximum deceleration speed of vehicle C_i, and δ is the driver reaction time of C_i to perform braking after seeing the braking light of vehicle C_{i-1} ($\delta = 1.5$ seconds in average [42]).

Based on the planned lane-changing space consisting of D_1, D_2,..., and D_q inter-vehicle gaps, we divide fleet vehicles into q groups for these consecutive gaps in the target lane. The required space for each fleet vehicle consists of its length and safety following distance, which is calculated as

$$A_i = b_{2-i} + E_{2-i}. \tag{5}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

After calculating the required spaces of all fleet vehicles, the number of fleet vehicles g_i that can move into each inter-vehicle gap D_i is calculated as

$$\begin{cases} \sum_{j=g_1}^{g_2-1} A_j \le D_1 < \sum_{j=g_1}^{g_2} A_j \\ \sum_{j=g_i}^{g_{i+1}-1} A_j \le D_i < \sum_{j=g_i}^{g_{i+1}} A_j \\ \sum_{j=g_q}^{n} A_j \le D_q \end{cases}, \tag{6}$$

where $g_1 = 1$, $i = 2, 3, \cdots, q-1$, and n is the total number of fleet vehicles. Note that if one or more fleet vehicles are not assigned to an inter-vehicle gap (due to gap fragmentation), the inter-vehicle gap (i.e., D_{q+1}) immediately following the planned lane-changing space is used for lane changing by the unassigned fleet vehicles and so forth.

For group speed control based on the planned lane-changing space, the expected colliding time $t_{2-g_j}^{dec}$ the first vehicle C_{g_j} in the j-th group would take to reach the incident location is calculated (considering the distance D_{emg} between the position X_{2-g_j} of the first vehicle in the j-th group and the location X_{emg} of the incident if the first vehicle in the j-th group maintains its current speed S_{2-g_j} as

$$\begin{cases} D_{emg} = X_{emg} - X_{2-g_j} \\ t_{2-g_j}^{dec} = \dfrac{D_{emg}}{S_{2-g_j}} \end{cases}. \tag{7}$$

As there is a sufficient aggregated lane-changing space, the distance between vehicles and the expected colliding time are used to calculate the required deceleration for each group of fleet vehicles, ensuring that the fleet groups can change lanes simultaneously without colliding with obstacles at the incident location. The deceleration for fleet vehicles in the j-th group to move into inter-vehicle gap D_i is calculated by solving the following simultaneous equations:

$$\begin{cases} Loc(D_i) - X_{2-g_j} - E'_{1-(i+1)} + (S_{1-i} - S'_{2-g_j})t^{cha} \\ \quad = (S_{2-g_j} - S_{1-i})t_{2-g_j}^{dec} - \dfrac{1}{2}a_{2-g_j}^{dec}(t_{2-g_j}^{dec})^2 \\ S'_{2-g_j} = S_{2-g_j} - a_{2-g_j}^{dec} t_{2-g_j}^{dec} \\ E'_{1-(i+1)} = \dfrac{(S'_{2-g_j})^2 - (S_{1-i})^2}{2a_{1-(i+1)}^{max}} + S'_{2-g_j}\delta \end{cases}, \tag{8}$$

where $Loc(D_i)$ is the starting location of inter-vehicle gap D_i, X_{2-g_j} is the location of the first vehicle in the j-th group, t^{cha} is the time taken for lane changing, $t_{2-g_j}^{dec}$ is the estimated time when the j-th group is expected to collide with obstacles at the incident location, $a_{2-g_j}^{dec}$ is the deceleration of fleet vehicles in the j-th group, S'_{2-g_j} is the speed of fleet vehicles in the j-th group after lane changing, and $E'_{1-(i+1)}$ is the safety following distance of target-lane vehicle $i+1$ after fleet lane changing.

Note that when the calculated deceleration of the rear group of fleet vehicles is less than that of the front group, there is a possibility of the rear group colliding with the front group. To prevent fleet group collisions, the following adjustment is performed as

$$a_{2-g_j}^{dec} = \max(a_{2-g_1}^{dec}, a_{2-g_2}^{dec}, \cdots, a_{2-g_j}^{dec}). \tag{9}$$

Algorithm 2 Dynamic Fleet Group Splitting

Input: S_{opt}, S_{start}, S_{end}, D^{gap}, A, N_F
Result: T, Dec

```
begin
    i = 1
    for j ← 1 to S_end − S_start + 1 do
        g_j^start = i
        while D^gap_{S_start+j} − A_i > 0 do
            D^gap_{S_start+j} = D^gap_{S_start+j} − A_i
            i = i + 1
    while i < N_F do
        D^gap_{S_end−S_start+1} = i
        while D^gap_{S_end+1} − A_i > 0 do
            D^gap_{S_end+1} = D^gap_{S_end+1} − A_i
            i = i + 1
        S_end = S_end + 1
    for k ← 1 to S_end − S_start do
        Calculate T_k and Dec_k based on Eq. (7), Eq. (8),
        and Eq. (9)
    return T, Dec
```

The deceleration of the current group of fleet vehicles is compared with the deceleration of the front group. When the deceleration of the current group is less than that of the front group, the deceleration of the current subgroup are set to the maximum value among the calculated decelerations of all the front groups. Algorithm 2 gives the pseudocode of the dynamic fleet group splitting algorithm as there is a sufficient aggregated lane-changing space. Through dynamic fleet group splitting,

C. Cooperative Fleet Group Splitting

To facilitate lane changing for a fleet of vehicles when there is no sufficient aggregated space, we further extend the design to coordinate the speeds of fleet vehicles and target-lane vehicles for cooperative fleet group splitting. When the number of selected inter-vehicle gaps is smaller, it can avoid a significant deceleration of fleet vehicles during lane changing. However, the aggregated space with a small number of inter-vehicle gaps may be far from the location of the lead vehicle in the fleet, which requires more time for lane changing with smaller deceleration. In contrast, if a larger number of inter-vehicle gaps is selected, more fleet vehicles can be accommodated, but fleet vehicles may need to perform a greater deceleration.

To achieve a balance between the number of selected gaps and the required deceleration, the spatial efficiency of consecutive inter-vehicle gaps is evaluated. The largest inter-vehicle gap D_{max} is used as a reference gap because D_{max} is the gap with the highest spatial efficiency that allows the most fleet vehicles to move into it. The spatial efficiency $P_{D(i,j)}$ of the aggregated lane-changing space $D(i, j)$ is calculated as

$$P_{D(i,j)} = \frac{D(i,j) - D_{max}}{L_F} \cdot \frac{1}{N(i,j)}, \tag{10}$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
SPATIAL EFFICIENCY OF AGGREGATED LANE-CHANGING SPACES (RELATIVE TO THE LARGEST VEHICLE GAP)

$P_{D(i,j)}$	1	2	3	4	5	6	7	8	9	10
1	-6.67	-1.00	1.56	2.00	2.80	3.56	3.90	4.42	4.96	5.53
2	-1.00	-6.00	0.33	1.33	2.50	3.47	3.89	4.48	5.08	5.70
3	1.56	0.33	-4.00	-0.33	1.78	3.17	3.73	4.44	5.14	5.83
4	2.00	1.33	-0.33	-7.33	-0.67	2.00	3.00	4.00	4.89	5.71
5	2.80	2.50	1.78	-0.67	-4.67	1.33	2.89	4.17	5.20	6.11
6	3.56	3.47	3.17	2.00	1.33	-3.33	1.33	3.56	5.00	6.13
7	3.90	3.89	3.73	3.00	2.89	1.33	-4.67	1.67	4.22	5.83
8	4.42	4.48	4.44	4.00	4.17	3.56	1.67	-2.67	3.33	5.78
9	4.96	5.08	5.14	4.89	5.20	5.00	4.22	3.33	-1.33	4.67
10	5.53	5.70	5.83	5.71	6.11	6.13	5.83	5.78	4.67	0.00

where L_F is the fleet length and $N(i, j)$ is the number of gaps from D_i to D_j.

For instance, assume that the fleet length increases from 52 meters to 150 meters, there is no sufficient aggregated space for the entire fleet of vehicles within a range of two times of the fleet length for lane changing. Tab. VI lists the spatial efficiency of each aggregated space in Tab. III, where the aggregated space with the highest efficiency is selected for lane changing. When the calculated spatial efficiency is negative, it means that the spatial efficiency of that aggregated space is lower than the referenced spatial efficiency of D_{max}, which is not eligible for selection (i.e., using the single gap D_{max} is better than the aggregated space). In Tab. VI, the selected aggregated space for lane changing is $D(6, 10)$ with the highest spatial efficiency. Once the aggregated space for lane changing is selected, fleet vehicles are allocated to each inter-vehicle gaps using Algorithm 2, whereas the last inter-vehicle gap is expanded due to insufficient space.

For space expansion, the expected colliding time for the last fleet group to reach the incident location is used to expand the required space through the deceleration of target-lane vehicles by considering the relationship between the deceleration and time for the last fleet group to decelerate to reach the last inter-vehicle gap, where the space required for the last fleet group consists of vehicle lengthes and safety distances between vehicles. The decelerations $a^{dec}_{2-g_j}$ and a^{dec}_{1-i} of the first fleet vehicle in the last group g_j and the i-th target-lane vehicle (located at the last inter-vehicle gap D_{i-1}) are calculated by solving the following simultaneous equations:

$$\begin{cases} h - D_{i-1} + 2E'_{(1-i)} = \\ \quad -\left((S_{1-i} - S_{1-(i-1)})t^{dec}_{2-g_j} + \frac{1}{2}a^{dec}_{1-i}(t^{dec}_{2-g_j})^2\right) \\ \sum_{k=g_j}^{n}(b_{2-k} + E'_{2-k}) = h \\ Loc(D_{i-1}) - X_{2-g_j} - E'_{1-i)} + (S'_{2-g_j} - S'_{1-i})t^{cha} = \\ \quad (S_{2-g_j} - S_{1-i})t^{dec}_{2-g_j} - \frac{1}{2}(a^{dec}_{2-g_j} - a^{dec}_{1-i})(t^{dec}_{2-g_j})^2 \end{cases}, \quad (11)$$

where h is the space required for lane changing of the last group, S'_{1-i} is the speed of the i-th vehicle in the target lane after fleet lane changing, and E'_{2-k} is the safety following distance of fleet vehicle k after lane changing. In particular, the decelerations of fleet group vehicles and target-lane vehicles are considered, both of which decelerate simultaneously. Algorithm 3 gives the pseudocode of the cooperative fleet group splitting algorithm as there is no sufficient aggregated lane-changing space. Through cooperative fleet group splitting, the speeds and acceleration/deceleration of fleet vehicles and target-lane vehicles are coordinated, which minimizes speed variations during lane changing.

Algorithm 3 Cooperative Fleet Group Splitting

Input: N_{gap}, D^{gap}, A
Result: S_{opt}, S_{start}, S_{end}, T, Dec

```
begin
    S_opt = 0
    for i ← 1 to N_gap do
        for j ← 1 to N_gap do
            P_D(i,j) = (D(i, j) − D_max)/(L_F · N(i, j))
            if S_opt < P_D(i,j) then
                S_opt = P_D(i,j)
                if i < j then
                    S_start = i
                    S_end = j
                else
                    S_start = j
                    S_end = i
    i = 1
    for j ← 1 to S_end − S_start do
        g_j^start = i
        while D^gap_(S_start+j) − A_i > 0 do
            D^gap_(S_start+j) = D^gap_(S_start+j) − A_i
            i = i + 1
    for k ← 1 to S_end − S_start + 1 do
        Calculate T_k, Dec_k^fleet, and Dec_k^target based on
        Eq. (7), Eq. (9), and Eq. (11)
    return S_opt, S_start, S_end, T, Dec
```

D. Multi-Group Fleet Merging

To merge fleet groups after passing the incident location, all group vehicles change to the original lane and form the complete fleet. The decelerations of fleet vehicles and target-lane vehicles are calculated with minimal speed variations and sufficient safety distances. In addition, the required times to expand space are determined as changing lanes to merge fleet groups in different inter-vehicle gaps. The sum of speed variations for affected vehicles (in the fleet lane and target lane) are estimated for merging space candidates, which is

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

used to determine which merging space is the optimal one with the minimum speed variation to merge fleet groups.

To avoid a secondary incident during the merging process, the location of the last fleet vehicle has to exceed the incident location with the safety following distance required at the current speed. The required decelerations and decelerating durations for fleet vehicles (in lane 1) and target-lane vehicles (in lane 2) to expand the merging space (i.e., inter-vehicle gap D_{k-1}) are calculated by solving the following simultaneous equations:

$$\begin{cases} (X_{1-1} - Loc(D_{2-(k-1)}) + b_{2-(k-1)}) + E_{2-(k-1)} = \\ \quad (S_{2-(k-1)} - S_{1-1})(t_{1-1}^{dec} - t^{cha}) + \frac{1}{2}a_{1-1}^{dec}(t_{1-1}^{dec} - t^{cha})^2 \\ (X_{1-1} - X_{1-n} + b_{1-n}) + E_{2-k} - D_{2-(k-1)} = \\ \quad (S_{2-(k-1)} - S_{2-k})(t_{1-1}^{dec} - t^{cha}) + \frac{1}{2}a_{2-k}^{dec}(t_{1-1}^{dec} - t^{cha})^2 \end{cases}, \tag{12}$$

where the time required for the first vehicle in the fleet to reach D_{k-1}, the decelerations of both fleet vehicles and target-lane vehicles, and the required space for merging the fleet based on the fleet position and safety distances are calculated.

Four fleet group merging methods are designed including Minimum Time Fleet Merging, Identical Deceleration Fleet Merging, Synchronous Velocity Fleet Merging, and Hybrid Measurement Fleet Merging:

(1) Minimum Time Fleet Merging: This method aims to minimize the deceleration time, allowing fleet vehicles to approach the target merging space as quickly as possible, which also minimizes the time required for target-lane vehicles to expand the merging space using the maximum braking deceleration as

$$a_{2-k}^{dec} = a_{2-k}^{max}. \tag{13}$$

Eq. (13) and Eq. (12) are simultaneously solved by substituting a_{2-k}^{dec} in the second equation with the maximum braking deceleration a_{2-k}^{max}, which can determine the time required for the fleet vehicles to reach the target merging space (i.e., t_{1-1}^{dec}) and the deceleration of the fleet vehicles (i.e., a_{1-1}^{dec}).

(2) Identical Deceleration Fleet Merging: This method uses the same deceleration for both fleet vehicles and target-lane vehicles to prevent the vehicle speeds from dropping two low. The decelerations required for fleet vehicles and target-lane vehicles and the time required for fleet merging are calculated using

$$a_{1-1}^{dec} = a_{2-k}^{dec}. \tag{14}$$

Eq. (14) and Eq. (12) are simultaneously solved by using the same deceleration for fleet vehicles and target-lane vehicles, which can determine the deceleration time t_{1-1}^{dec}, the deceleration speed a_{1-1}^{dec}, and the deceleration of the target-lane vehicle (i.e., a_{2-k}^{dec}).

(3) Synchronous Velocity Fleet Merging: This method synchronizes the speed of fleet vehicles with the deceleration of target-lane vehicles to avoid excessive deceleration of affected vehicles. During the merging of fleet groups, the final speed of fleet vehicles matches that of target-lane vehicles using

$$S_{1-1} - a_{1-1}^{dec}(t_{1-1}^{dec} - t^{cha}) = S_{2-k} - a_{2-k}^{dec}(t_{1-1}^{dec} - t^{cha}). \tag{15}$$

Eq. (15) and Eq. (12) are simultaneously solved to ensure that the speeds of fleet vehicles and target-lane vehicles are the same after expanding the target merging space.

(4) Hybrid Measurement Fleet Merging: On one hand, Minimum Time Fleet Merging uses the maximum deceleration of vehicles to merge fleet groups, leading to excessive deceleration in the target lane and causing a significant speed change. On the other hand, when the speed of target-lane vehicles is less than the fleet speed, Identical Deceleration Fleet Merging results in lower speed variations than Synchronous Velocity Fleet Merging. However, when the speed of target-lane vehicles is greater than the fleet speed, Synchronous Velocity Fleet Merging outperforms Identical Deceleration Fleet Merging. To address the differences caused by vehicle speeds, Hybrid Measurement Fleet Merging is designed to simultaneously estimate the speed variations using Identical Deceleration and Synchronous Velocity merging methods and select one of them with the minimum speed variation for merging fleet groups, which minimizes unnecessary acceleration/deceleration during the merging process, thereby reducing fuel/energy consumption.

After calculating the required decelerations for fleet vehicles and target-lane vehicles, the speed variation caused by merging fleet groups in each merging space behind the lead vehicle in the fleet are estimated. The total speed variation of affected vehicles (i.e., Q) are calculated using fleet vehicle speed variation (i.e., Q_{fleet}) and target-lane vehicle speed variation (i.e., Q_{target}) as

$$Q = Q_{fleet} + Q_{target}. \tag{16}$$

The fleet vehicle speed variation consists of two parts. The first part is the speed variation when fleet vehicles decelerate to the target merging space, and the second part is the speed variation required when the merged fleet vehicles follow the front target-lane vehicle with the same speed, which is calculated as

$$Q_{fleet} = |a_{1-1}^{dec} t_{1-1}^{dec}| + |S_{2-(k-1)} - (S_{1-1} - a_{1-1}^{dec} t_{1-1}^{dec})|. \tag{17}$$

The target-lane vehicle performs deceleration to expand the merging space, and when fleet vehicles merge, it returns to its original speed. Therefore, the target-lane vehicle speed variation is calculated as

$$Q_{target} = 2 \cdot |a_{2-k}^{dec} t_{1-1}^{dec}|. \tag{18}$$

Algorithm 4 gives the pseudocode of the multi-group fleet merging algorithm, where I_{ID} and I_{SV} are the merging spaces selected by Identical Deceleration and Synchronous Velocity merging methods, respectively. Through multi-group fleet merging, the feasible space required for merging fleet groups is planned based on the number of fleet groups and their positions, which achieves fleet merging with minimal speed variations.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

Algorithm 4 Multi-Group Fleet Merging

Input: N_{gap}
Result: T, Dec, I
begin
  $I^{ID} = 1$
  $I^{SV} = 1$
  for $i \leftarrow 1$ **to** N_{gap} **do**
    Calculate Dec_i^{ID}, Dec_{fleet}^{ID}, and T_i^{ID} based on Eq. (12) and Eq. (14)
    Calculate Q_{fleet}^{ID} and Q_{target}^{ID} based on Eq. (17) and Eq. (18)
    $Q_i^{ID} = Q_{fleet}^{ID} + Q_{target}^{ID}$
    if $Q_i^{ID} < Q_{i-1}^{ID} \& i > 1$ **then**
      $I^{ID} = i$
      $T^{ID} = T_i^{ID}$
    Calculate Dec_i^{SV}, Dec_{fleet}^{SV}, and T_i^{SV} based on Eq. (12) and Eq. (15)
    Calculate Q_{fleet}^{SV} and Q_{target}^{SV} based on Eq. (17) and Eq. (18)
    $Q_i^{SV} = Q_{fleet}^{SV} + Q_{target}^{SV}$
    if $Q_i^{SV} < Q_{i-1}^{SV}$ & $i > 1$ **then**
      $I^{SV} = i$
      $T^{SV} = T_i^{SV}$
  if $min(Q^{ID}) < min(Q^{SV})$ **then**
    return T^{ID}, Dec^{ID}, I^{ID}
  else
    return T^{SV}, Dec^{SV}, I^{SV}

TABLE VII
SIMULATION PARAMETERS

Parameter	Value
Fleet Size	5 ∼ 25 vehicles
Vehicle Speed	16 ∼ 30 m/s
Initial Distance	35 ∼ 60 m
Maximum Acceleration	6 m/s^2
Maximum Deceleration	8 m/s^2
Driver Reaction Time	1.5 s
Communication Range	300 m
Lane-Changing Time	3 s
Lane Width	4 m

V. PERFORMANCE EVALUATION

In this section, we evaluate the performance of the proposed framework by comparing it with existing methods. The simulation tool used for experiments is SUMO (Simulation of Urban MObility) with CACC car following model [43], which is an open-source microscopic traffic simulation software. Fleets with 5, 10, $\cdots$, and 25 vehicles in the fleet lane (i.e., lane 2) and different numbers of non-fleet vehicles in the target lane (i.e., lane 1) are simulated, where the initial distance between vehicles is randomly assigned from 35 m to 60 m and the default communication range of each vehicle is 300 m. Tab. VII lists the parameters used in the experiments, including fleet size, vehicle speed, maximum deceleration, maximum acceleration, driver reaction time, lane-changing time, and lane width.

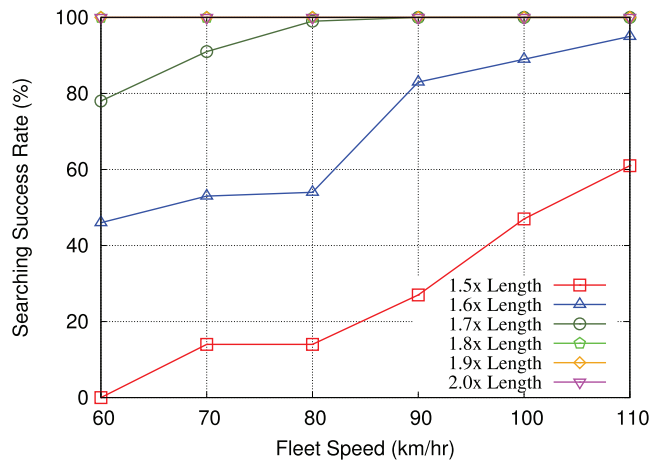

Fig. 5. Comparison of searching success rates to find sufficient aggregated lane-changing spaces.

The speeds of fleet vehicles are set from 60 km/hr to 110 km/hr (i.e., around 16 m/s ∼ 30 m/s) and their maximum acceleration and deceleration are 6 m/s^2 and 8 m/s^2, respectively.

The speed variation for a fleet vehicle is the sum of the decreasing speed from the original speed to the lane-changing speed (in the fleet lane) and the increasing speed from the lane-changing speed to the lane speed (in the target lane), where the average speed variation is the sum of speed variations for affected vehicles (in the fleet lane and target lane) divided by the number of affected vehicles. In the following experiments, we compare searching success rates of sufficient aggregated lane-changing spaces under different vehicle speeds. In addition, we compare different lane changing approaches, evaluating average speed variations caused by existing methods and our framework. Furthermore, we compare average speed variations required for fleet merging using four designed methods.

First, we conduct experiments to compare different searching ranges for finding sufficient aggregated lane-changing spaces, where the searching range is α times of the fleet length. When the vehicles are moving at higher speeds, a larger safety following distance between vehicles is required. We aim to find the suitable value of α for aggregated lane-changing space searching with varying fleet speeds. With a fleet of 10 vehicles, a searching range of 1.0 times of the fleet length cannot provide sufficient aggregated lane-changing spaces. Fig. 5 shows that even with a searching range of 1.7 times of the fleet length, with lower vehicle speeds, there is still a chance of not finding sufficient aggregated lane-changing spaces. In contrast, with a searching range of 2.0 times of the fleet length, sufficient aggregated lane-changing spaces can be found under all fleet speeds. However, with larger searching ranges, the found aggregated lane-changing spaces may be far from the fleet, requiring significant deceleration for fleet vehicles to reach the target aggregated lane-changing space. This results in a significant increase in speed variations for fleet vehicles before and after lane changing, leading to higher fuel consumption and a substantial reduction in traffic flow smoothness. Therefore, we use 2 times of the fleet length as

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

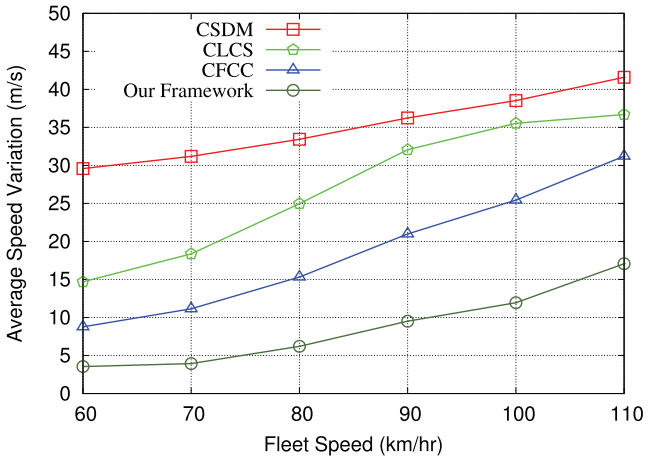

Fig. 6. Comparison of average speed variations of affected vehicles under different fleet speeds with sufficient aggregated lang-changing spaces.

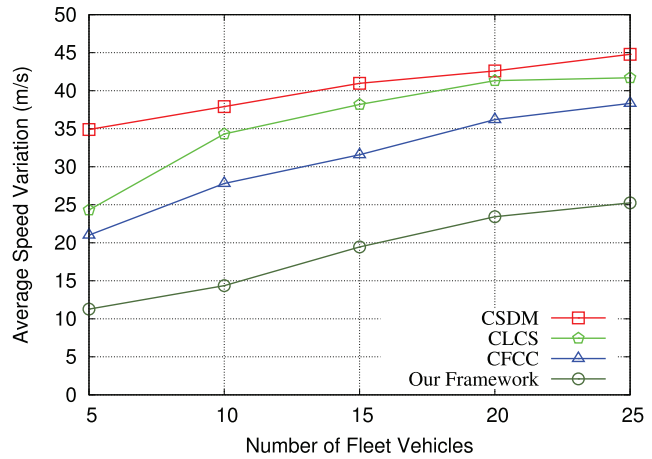

Fig. 7. Comparison of average speed variations of affected vehicles under different numbers of fleet vehicles with sufficient aggregated lang-changing spaces.

our designated searching range. If sufficient aggregated lane-changing spaces are still not found within this range (using Algorithm 2), cooperative adjustments of the speeds between fleet vehicles and target-lane vehicles (using Algorithm 3) can be employed to expand the space to accommodate all fleet vehicles successfully.

Next, we conduct experiments to compare average speed variations of lane changing using the proposed framework and existing methods, including CSDM [35], CLCS [36], and CFCC [37]. In CSDM [35], the positions, speeds, and spaces of adjacent vehicles are considered. When a vehicle changes its lane, the rear vehicle in the target lane assesses the required safety following distance based on the speed of the front vehicle and the driver reaction time. If the current following distance is less than the required safety distance, the rear vehicle decelerates. Thus, lane-changing space expansion enables cooperative lane changing for vehicles in the target lane. In CLCS [36], it allows coordinated lane changing based on the relative locations of vehicles in the current and adjacent lanes. CLCS addresses road safety and congestion issues but focuses on individual vehicles, leading to a lane-changing process conducted separately for each vehicle. In CFCC [37], it calculates risk values based on the speeds of the fleet vehicles and the preceding vehicle, adjusting the optimal following speed according to the risk value. When an accident occurs in front of the fleet, CFCC estimates the expected colliding time and performs fleet lane changing without multi-group splitting and merging.

Fig. 6 shows the comparison of average speed variations of affected vehicles (i.e., fleet vehicles and target-lane vehicles) under the fleet speeds of 60, 70, $\cdots$, and 110 km/hr with sufficient aggregated lang-changing spaces (for a fleet of 10 vehicles). It can be observed that our framework consistently achieves the minimum average speed variation when performing fleet lane-changing actions. In particular, at higher speeds, the difference between our framework and existing methods becomes larger, which accomplishes fleet lane-changing with less speed variation. In particular, with a fleet of 10 vehicles at 110 km/hr, average speed variations for changing lanes are decreased by more than 14 m/s and improved by more than 45%. This is because our fleet group splitting does not require the vehicles in the target lane to decelerate when there is no single sufficient lane-changing space. Therefore, our framework avoids unnecessary deceleration of target-lane vehicles, lowering the average speed variations of affected vehicles.

Fig. 7 shows the comparison of average speed variations of affected vehicles when the fleet changes its lane with 5, 10, $\cdots$, and 25 fleet vehicles, where the speed of fleet vehicles is 90 km/hr and the speed of target-lane vehicles is 110 km/hr. It can be seen that our framework achieves the minimum average speed variation under all fleet sizes during lane changing. This is because our framework requires only the fleet to decelerate, with no need for the vehicles in the target lane to slow down for the expansion of space. Existing methods use a single inter-vehicle gap to perform fleet lane changing, and since a single gap requires significant space expansion to accommodate all fleet vehicles, the vehicles in the target lane experience substantial deceleration. Therefore, our framework allows the fleet for lane changing with the minimum speed variation.

Fig. 8 shows the comparison of average speed variations among vehicles when a 10-vehicle fleet changes its lane, and there is no sufficient aggregated space in the target lane. Our framework, even in situations where the aggregated space is insufficient, achieves the minimum average speed variation for lane-changing actions regardless of the fleet speeds. Moreover, as the fleet speed increases, the required safety distance also increases, necessitating more space to accommodate all fleet vehicles for lane changing. In our framework, we only need to expand the last gap in the aggregated space for the fleet vehicles in the last group. In contrast, existing methods rely on a single inter-vehicle gap for lane changing, where the required expansion space is more significant. The vehicles in the target lane need to decelerate significantly to expand space for enabling fleet vehicles to change lanes successfully.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

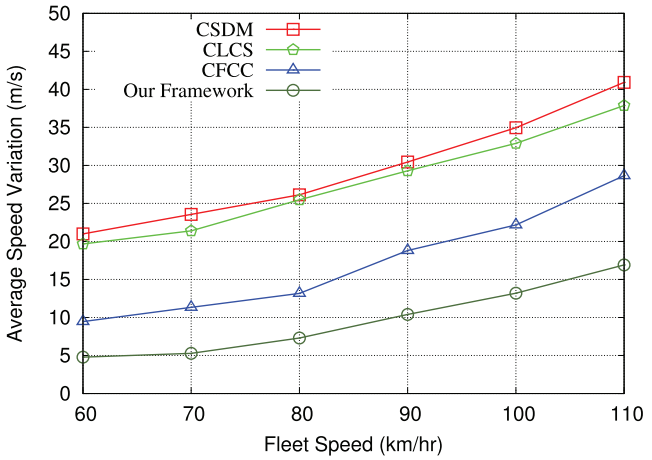

Fig. 8. Comparison of average speed variations of affected vehicles under different fleet speeds with insufficient aggregated lang-changing spaces.

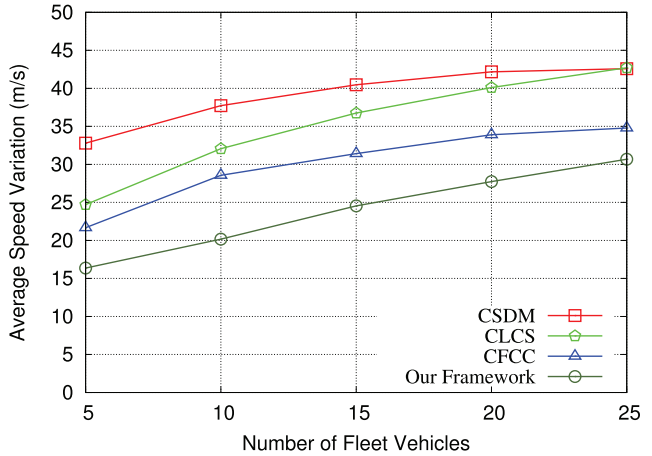

Fig. 9. Comparison of average speed variations of affected vehicles under different numbers of fleet vehicles with insufficient aggregated lang-changing spaces.

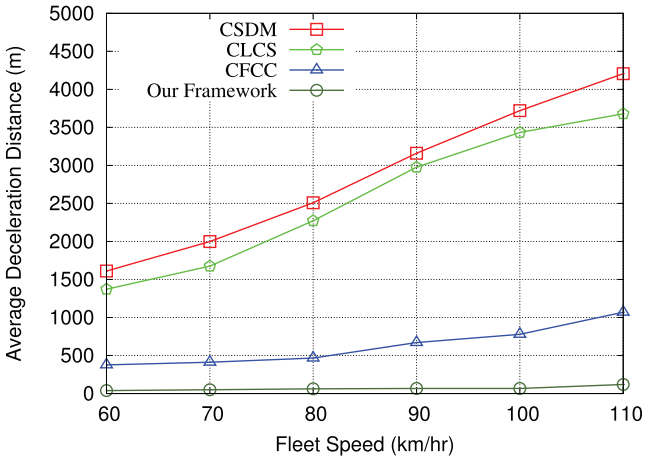

Fig. 10. Comparison of average deceleration distances of target-lane vehicles under different fleet speeds with insufficient aggregated lang-changing spaces.

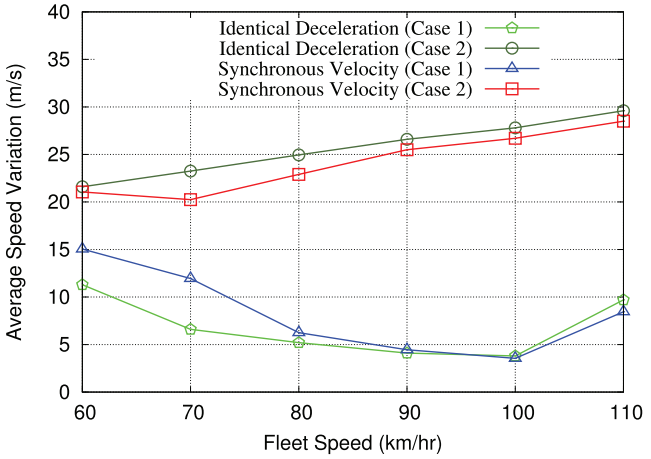

Fig. 11. Comparison of average speed variations between Identical Deceleration Fleet Merging and Synchronous Velocity Fleet Merging in different cases.

Fig. 9 shows the comparison of average speed variations among vehicles when the fleet changes its lane with varying fleet sizes and insufficient aggregated spaces, where the speed of fleet vehicles is 90 km/hr and the speed of vehicles in the target lane is 110 km/hr. When there is no sufficient aggregated space, the average speed variation trends similarly in our framework with sufficient aggregated spaces. This is because our framework only needs to expand the last gap of the aggregated space. Existing methods, considering only a single inter-vehicle gap closest to the fleet vehicles, experience a higher increase in average speed variations when the gap space is insufficient. In particular, with a larger number of fleet vehicles, our framework shows a trend of converging average speed variations with existing methods, as it requires the splitting of more groups.

Fig. 10 shows the difference in average deceleration distances between target-lane vehicles with braking (i.e., some deceleration distance) and without braking (i.e., no deceleration distance), which aims to evaluate the impact of fleet vehicles changing lanes on the deceleration of vehicles in the target lane. As shown in Fig. 10, our framework only affects the vehicles behind the last inter-vehicle gap of the aggregated space. In contrast, existing methods only select the inter-vehicle gap closest to fleet vehicles for expanding, resulting in a significant difference in average deceleration distances with more affected target-lane vehicles and longer deceleration distances.

Finally, we conduct experiments to evaluate four designed merging methods: Minimum Time, Identical Deceleration, Synchronous Velocity, and Hybrid Measurement Fleet Merging. Using the method that minimizes average speed variations during fleet merging, our framework aim to reduce fuel/energy consumption and effectively decrease the impact on traffic flow smoothness. Fig. 11 shows the comparison of average speed variations between Identical Deceleration Fleet Merging and Synchronous Velocity Fleet Merging with the fleet speeds of 60, 70, $\cdots$, and 110 km/hr in different cases (i.e., Case 1 and Case 2), where the fleet consists of 10 vehicles, divided into three groups, each separated by a non-fleet vehicle.

In Case 1, the speed of target-lane vehicles is 110 km/hr, whereas in Case 2, the speed of target-lane vehicles

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

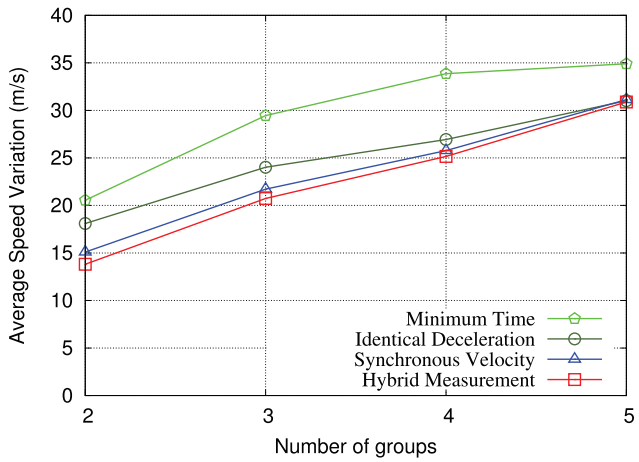

Fig. 12. Comparison of average speed variations during fleet merging for different numbers of groups and different merging methods.

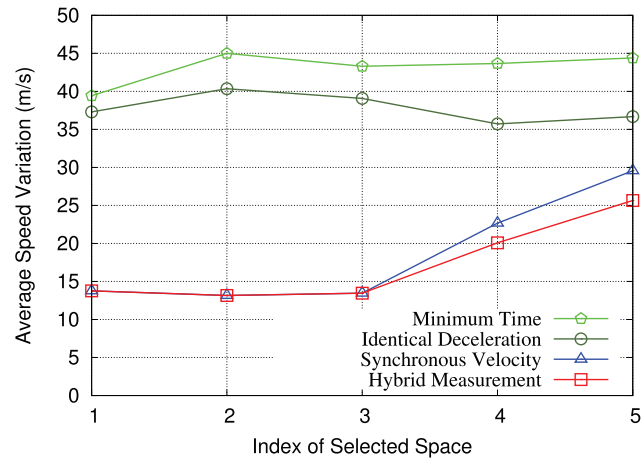

Fig. 13. Comparison of average speed variations as selecting different merging spaces for fleet merging.

is 60 km/hr. As shown in Fig. 11, both merging methods have respective advantages and disadvantages in different cases. Identical Deceleration Fleet Merging often has a lower average speed variation than Synchronous Velocity Fleet Merging when the fleet speed is lower than the target-lane vehicle speed, and the selected merging space does not require much expansion. However, it loses its advantage when the speeds of the two lanes are similar because the fleet vehicles need significant deceleration to reach the selected merging space, resulting in higher average speed variations. Synchronous Velocity Fleet Merging performs better when the speeds of the two lanes are close because it can use a smaller deceleration to slow down the fleet vehicles and expand the selected merging space in the target lane. Therefore, we combine the two merging methods into Hybrid Measurement Fleet Merging, leveraging the advantages of both methods in different cases, further reducing the average speed variation during fleet merging.

Fig. 12 shows the comparison of average speed variations during fleet merging for different numbers of groups using four designed merging methods, where the fleet consists of 10 vehicles and each group is separated by a non-fleet vehicle. It can be seen that Hybrid Measurement Fleet Merging outperforms other merging methods and achieves the lowest average speed variation. As the number of groups in the fleet splitting increases, the total length of the fleet becomes longer, requiring a larger space for fleet merging. As shown in Fig. 12, when there are more fleet groups, the target-lane vehicles need to decelerate more to provide enough space for fleet groups to merge.

Fig. 13 shows the comparison of average speed variations as selecting different merging spaces for fleet merging. The fleet consists of 10 vehicles, which is divided into three groups and each group is separated by a non-fleet vehicle. The fleet speed is 60 km/hr and the speed of target-lane vehicles is 110 km/hr. The smaller the index value of the merging space, the closer it is to the lead vehicle in the fleet. The basis for selecting spaces for fleet merging, as shown in Fig. 13, requires considering both spatial position and space size, where the merging space

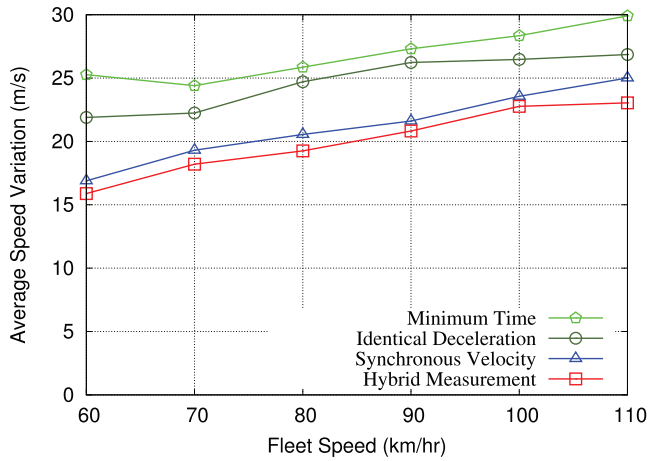

Fig. 14. Comparison of average speed variations using four merging methods under different fleet speeds.

closest to the lead vehicle in the fleet does not always achieves the lowest average speed variation among all merging spaces.

Fig. 14 shows the comparison of average speed variations using four merging methods under different fleet speeds, with a fleet of 10 vehicles divided into three subgroups, each separated by a non-fleet vehicle. It can be seen that the average speed variation for Minimum Time Fleet Merge is the highest under all fleet speeds. This is because the target-lane vehicle uses the maximum deceleration to expand the space, and the fleet vehicles also need to decelerate significantly to reach the selected merging space, resulting in the highest average speed variation for this merging method. On the other hand, the average speed variation for Synchronous Velocity Fleet Merging is lower than both Minimum Time and Identical Deceleration Fleet Merging. In addition, Hybrid Measurement Fleet Merging combining Identical Deceleration and Synchronous Velocity merging methods minimizes the speed variation of merged fleet vehicles. In particular, all four merging methods show an increase in average speed variations due to the speed rising of the fleet. This is because each fleet group has non-fleet vehicles between them, which requires a

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

larger safety distance and more space for fleet group merging, leading to an increase in average speed variations.

VI. Conclusion

In this paper, we propose a framework for fleet group splitting and merging in cooperative fleet lane changing. We design a dynamic programming algorithm for lane-changing space optimization. In addition, as aggregated lane-changing spaces are insufficient, our framework provides cooperative speed and space adjustments to expand the aggregated lane-changing space, facilitating the seamless lane-changing of fleet vehicles. Furthermore, we design four methods for fleet merging after passing through the incident location, which allow fleet groups to change to the original lane and form the complete fleet. Throughout the process of fleet splitting and group merging, sufficient following distances are maintained between vehicles for safety. Our framework minimizes the speed variation of fleet vehicles during lane changing and group merging, which avoids unnecessary acceleration and deceleration to reduce fuel/energy consumption. For future works, we are exploring the lane-changing space aggregated from both adjacent lanes of the current fleet lane, where fleet groups can simultaneously move into the inter-vehicle gaps on both adjacent lanes, allowing for a faster and more efficient fleet lane changing. In addition, real-world experiments will be conducted to further verify the effectiveness and practicality of the proposed framework.

References

[1] U. Montanaro, M. Tufo, G. Fiengo, M. Di Bernardo, A. Salvi, and S. Santini, "Extended cooperative adaptive cruise control," in *Proc. IEEE Intell. Vehicles Symp.*, Jun. 2014, pp. 605–610.

[2] X. Zheng, S. Li, X. Luo, Y. Zhang, X. Li, and X. Guan, "Fast distributed platooning of connected vehicular systems with inaccurate velocity measurement," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 10, pp. 5996–6006, Oct. 2023.

[3] X. Liao et al., "Game theory-based ramp merging for mixed traffic with unity-SUMO co-simulation," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 9, pp. 5746–5757, Sep. 2022.

[4] S. Gyawali, S. Xu, Y. Qian, and R. Q. Hu, "Challenges and solutions for cellular based V2X communications," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 222–255, 1st Quart., 2021.

[5] D. Pan, D. Ding, X. Ge, Q.-L. Han, and X.-M. Zhang, "Privacy-preserving platooning control of vehicular cyber–physical systems with saturated inputs," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 4, pp. 2083–2097, Apr. 2023.

[6] C. Jauregui, M. Torres, M. Silvera, and F. Campos, "Improving people's accessibility through a fully actuated signal control at intersections with high density of pedestrians," in *Proc. Congreso Internacional de Innovación y Tendencias en Innovación (CONIITI)*, Sep. 2020, pp. 1–6.

[7] N. C. Maduka, I. I. Ajibade, and M. Bello, "Modelling and optimization of smart traffic light control system," in *Proc. IEEE Nigeria 4th Int. Conf. Disruptive Technol. Sustain. Develop. (NIGERCON)*, Apr. 2022, pp. 1–5.

[8] B. Li et al., "Sharing traffic priorities via cyber–physical–social intelligence: A lane-free autonomous intersection management method in metaverse," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 4, pp. 2025–2036, Apr. 2023.

[9] Y. Ni, C. Zhao, and L. Cai, "Hybrid RSU management in cybertwin-IoV for temporal and spatial service coverage," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 4596–4606, May 2022.

[10] J. Zhou and H. Peng, "Range policy of adaptive cruise control vehicles for improved flow stability and string stability," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 2, pp. 229–237, Jun. 2005.

[11] A. Langer and S. McRae, "Step on it: A new approach to improving vehicle fuel economy," Dept. Econ., Univ. Arizona, Tucson, AZ, USA, Tech. Rep., Jan. 2017.

[12] A. Langer and S. McRae, "Step on it: Approaches to improving existing vehicles' fuel economy," Dept. Econ., Univ. Arizona, Tucson, AZ, USA, Tech. Rep., Dec. 2014.

[13] V. Milanés, S. E. Shladover, J. Spring, C. Nowakowski, H. Kawazoe, and M. Nakamura, "Cooperative adaptive cruise control in real traffic situations," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 1, pp. 296–305, Feb. 2014.

[14] F. Navas and V. Milanés, "Mixing V2V{-} and non-V2V-equipped vehicles in car following," *Transp. Res. C, Emerg. Technol.*, vol. 108, pp. 167–181, Nov. 2019.

[15] H. Liu, X.-Y. Lu, and S. E. Shladover, "Traffic signal control by leveraging cooperative adaptive cruise control (CACC) vehicle platooning capabilities," *Transp. Res. C, Emerg. Technol.*, vol. 104, pp. 390–407, Jul. 2019.

[16] C. L. Melson, M. W. Levin, B. E. Hammit, and S. D. Boyles, "Dynamic traffic assignment of cooperative adaptive cruise control," *Transp. Res. C, Emerg. Technol.*, vol. 90, pp. 114–133, May 2018.

[17] D. Xiaoping, L. Dongxin, L. Shen, W. Qiqige, and C. Wenbo, "Coordinated control algorithm at non-recurrent freeway bottlenecks for intelligent and connected vehicles," *IEEE Access*, vol. 8, pp. 51621–51633, 2020.

[18] R. K. Kamalanathsharma and H. Rakha, "Agent-based modeling of eco-cooperative adaptive cruise control systems in the vicinity of intersections," in *Proc. 15th Int. IEEE Conf. Intell. Transp. Syst.*, Sep. 2012, pp. 840–845.

[19] H. Yang, H. Rakha, and M. V. Ala, "Eco-cooperative adaptive cruise control at signalized intersections considering queue effects," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 6, pp. 1575–1585, Jun. 2017.

[20] M. H. Almannaa, H. Chen, H. A. Rakha, A. Loulizi, and I. El-Shawarby, "Field implementation and testing of an automated eco-cooperative adaptive cruise control system in the vicinity of signalized intersections," *Transp. Res. D, Transp. Environ.*, vol. 67, pp. 244–262, Feb. 2019.

[21] R. A. Shet and F. Schewe, "Performance evaluation of cruise controls and their impact on passenger comfort in autonomous vehicle platoons," in *Proc. IEEE 89th Veh. Technol. Conf. (VTC-Spring)*, Apr. 2019, pp. 1–7.

[22] S. Tajeddin, S. Ekhtiari, M. Faieghi, and N. L. Azad, "Ecological adaptive cruise control with optimal lane selection in connected vehicle environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 11, pp. 4538–4549, Nov. 2020.

[23] J. S. Brunner, M. A. Makridis, and A. Kouvelas, "Comparing the observable response times of ACC and CACC systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19299–19308, Oct. 2022.

[24] G. Xu, J. Wang, G. Q. Huang, and C.-H. Chen, "Data-driven resilient fleet management for cloud asset-enabled urban flood control," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 6, pp. 1827–1838, Jun. 2018.

[25] R. Luo, T. J. J. Van Den Boom, and B. De Schutter, "Multi-agent dynamic routing of a fleet of cybercars," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 5, pp. 1340–1352, May 2018.

[26] S. Mehar, S. Zeadally, G. Rémy, and S. M. Senouci, "Sustainable transportation management system for a fleet of electric vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 3, pp. 1401–1414, Jun. 2015.

[27] Z. Yi, J. Smart, and M. Shirk, "Energy impact evaluation for eco-routing and charging of autonomous electric vehicle fleet: Ambient temperature consideration," *Transp. Res. C, Emerg. Technol.*, vol. 89, pp. 344–363, Apr. 2018.

[28] J. Suh, H. Chae, and K. Yi, "Stochastic model-predictive control for lane change decision of automated driving vehicles," *IEEE Trans. Veh. Technol.*, vol. 67, no. 6, pp. 4771–4782, Jun. 2018.

[29] Y. Zhang and P. A. Ioannou, "Combined variable speed limit and lane change control for highway traffic," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 7, pp. 1812–1823, Jul. 2017.

[30] N. Wiesner, J. Sheppard, and B. Haberman, "Using particle swarm optimization to learn a lane change model for autonomous vehicle merging," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2021, pp. 1–8.

[31] Y. Guo, H. Xu, Y. Zhang, and D. Yao, "Integrated variable speed limits and lane-changing control for freeway lane-drop bottlenecks," *IEEE Access*, vol. 8, pp. 54710–54721, 2020.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.

[32] Y. He, D. Sun, M. Zhao, and S. Cheng, "Cooperative driving and lane changing modeling for connected vehicles in the vicinity of traffic signals: A cyber-physical perspective," *IEEE Access*, vol. 6, pp. 13891–13897, 2018.

[33] B. Li, Y. Zhang, Y. Feng, Y. Zhang, Y. Ge, and Z. Shao, "Balancing computation speed and quality: A decentralized motion planning method for cooperative lane changes of connected and automated vehicles," *IEEE Trans. Intell. Vehicles*, vol. 3, no. 3, pp. 340–350, Sep. 2018.

[34] M. Sun, Z. Chen, H. Li, and B. Fu, "Cooperative lane-changing strategy for intelligent vehicles," in *Proc. 40th Chin. Control Conf. (CCC)*, Jul. 2021, pp. 6022–6027.

[35] T. T. Phan, L. B. Le, and D. Ngoduy, "A cooperative space distribution method for autonomous vehicles at a lane-drop bottleneck on multi-lane freeways," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3710–3723, Apr. 2022.

[36] M. Yang, C. Li, W. Wu, and C. Qi, "A cooperative lane change strategy for improving road safety through V2V communications," in *Proc. 13th Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, Oct. 2022, pp. 1415–1418.

[37] L.-W. Chen and G.-L. Wang, "Risk-aware and collision-preventive cooperative fleet cruise control based on vehicular sensor networks," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 1, pp. 179–191, Jan. 2022.

[38] Y. Jiang, L. Tan, G. Xiao, Y. Wu, and Z. Yao, "Platoon-aware cooperative lane-changing strategy for connected automated vehicles in mixed traffic flow," *Phys. A, Stat. Mech. Appl.*, vol. 640, pp. 1–23, Apr. 2024.

[39] J. Zhang, C. Chang, X. Zeng, and L. Li, "Multi-agent DRL-based lane change with right-of-way collaboration awareness," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 854–869, Jan. 2023.

[40] B. Li, X. Dong, and J. Wen, "Cooperative-driving control for mixed fleets at wireless charging sections for lane changing behaviour," *Energy*, vol. 243, pp. 1–14, Mar. 2022.

[41] C. Zhao, Z. Li, L. Li, X. Wu, and F.-Y. Wang, "A negotiation-based right-of-way assignment strategy to ensure traffic safety and efficiency in lane changes," *IET Intell. Transp. Syst.*, vol. 15, no. 11, pp. 1345–1358, Nov. 2021.

[42] N. Lerner, "Brake perception-reaction times of older and younger drivers," in *Proc. Human Factors Ergonom. Soc. Annu. Meeting*, vol. 37, Oct. 1993, pp. 206–210.

[43] P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in *Proc. 21st Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2018, pp. 2575–2582.

Lien-Wu Chen (Senior Member, IEEE) received the Ph.D. degree in computer science and information engineering from the National Chiao Tung University in December 2008.

He joined the Department of Computer Science, National Chiao Tung University as a Post-Doctoral Research Associate in February 2009, and qualified as an Assistant Research Fellow in February 2010. From 2012 to 2015, he was an Assistant Professor with the Department of Information Engineering and Computer Science, Feng Chia University, where he was the Chief of Operation and Maintenance Section, from 2016 to 2018. He was an Associate Professor from 2015 to 2019, and he has been a Full Professor, since February 2019. He has published over 150 journal and conference papers and holds over 20 granted invention patents. His research interests include wireless communication and mobile computing, especially in the Internet of Vehicles, the Internet of Things, and the Internet of People.

Dr. Chen received over 30 Best/Excellent Paper Awards, the Best/Outstanding Demo Awards, the Best Presentation Awards, and the Outstanding Paper Publication Awards, since 2009, and transferred three technologies to Acer Incorporated in January, June, and November 2011. He is the advisor of over 60 Prototyping Awards, the Student Engineering Paper Awards, the College Student Research Awards, and the Master Thesis Awards since 2014. He is a Honorary Member of Phi Tau Phi Scholastic Honor Society, a Lifetime Member of Chinese Institute of Electrical Engineering, the Institute of Information and Computing Machinery, and Taiwan Institute of Electrical and Electronic Engineering, and a Senior Member of ACM.

Yu-An Shi received the B.S. and M.S. degrees in information engineering and computer science from Feng Chia University, Taichung, Taiwan, in 2020 and 2023, respectively. His research interests include the Internet of Things and intelligent transportation systems. He was a recipient of the Best Paper Award with the Workshop on Wireless, Ad Hoc, and Sensor Networks in 2024.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 02,2026 at 07:21:41 UTC from IEEE Xplore. Restrictions apply.